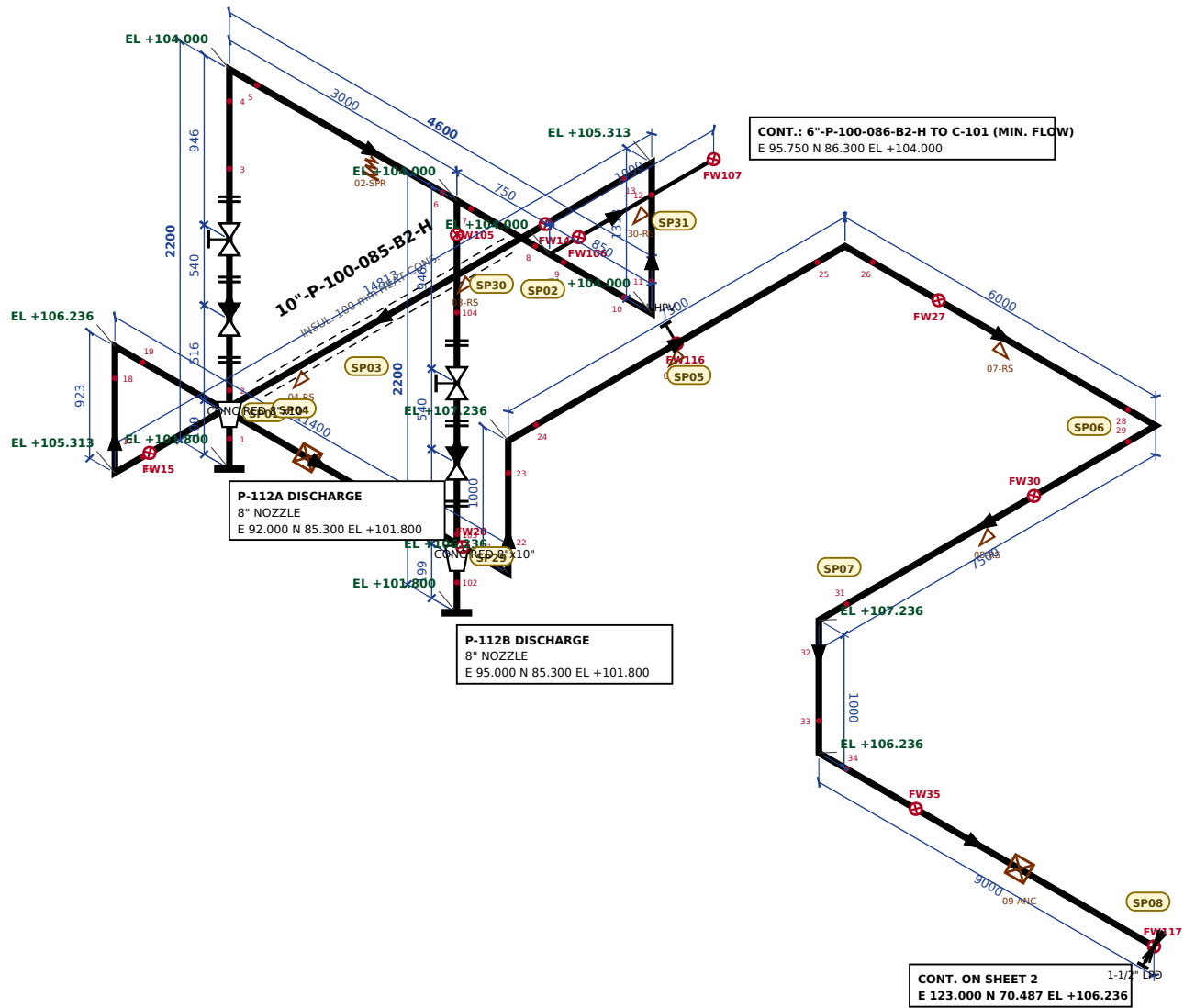
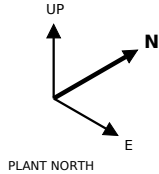


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	8"	2
2	REDUCER CONC. BW, SCH 40, ASTM A234 WP5	10" x 8"	2
3	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	10"	4
4	PIPE SCH 40, BE, ASTM A335 P5 SMLS	10"	62.1 m
5	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP5	10"	11
6	TEE EQUAL, BW, SCH 40, ASTM A234 WP5	10"	1
7	TEE RED., BW, SCH 40, ASTM A234 WP5	10" x 6"	1
8	PIPE SCH 40, BE, ASTM A335 P5 SMLS	6"	0.8 m
9	SOCKOLET 3000# (branch), ASTM A182 F5	1"	1
10	SOCKOLET 3000# (branch), ASTM A182 F5	1-1/2"	1
11	ERECTION: CHECK VALVE SWING CL300 RF, A217 C5, trim 5	10"	2
12	GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	10"	2
13	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	10"	6
14	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	10"	6
15	PIPE SUPPORTS (std. dwgs): anchor x2, rest x6, spring x1	-	9

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP02-1	10"	220	SP02-2	10"	2403	SP02-3	10"	318
SP02-4	10"	253	SP02-5	10"	551	SP02-6	10"	2250
SP03-1	10"	11600	SP04-1	10"	201	SP04-2	10"	161
SP04-3	10"	10050	SP05-1	10"	588	SP05-2	10"	238
SP05-3	10"	6738	SP05-4	10"	1350	SP06-1	10"	3888
SP06-2	10"	2250	SP07-1	10"	4488	SP07-2	10"	238
SP07-3	10"	2250	SP08-1	10"	11700	SP30-1	10"	385
SP31-1	6"	784						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=0.4m SP02=9.3m SP03=11.6m SP04=11.9m SP05=11.2m SP06=6.9m SP07=8.5m SP08=11.7m SP29=0.4m SP30=0.5m SP31=0.8m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 44.7 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Discharge check + block valves in riser above each pump (accessible from grade).
- Rack run on tier 1 (hot B2) at rack south edge; expansion loops raised +1.0 m.
- FV-1083 / FE / XV station at grade (EL 101.2) north of H-201 for operator access.
- Riser at H-201 supported from heater structure with variable spring hangers.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 10"-P-100-085-B2-H

P-112A/B DISCHARGE TO H-201 INLET

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 3
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER	CFU-100-PI-ISO-004		
			REV
			A

LINE NUMBER	FROM	TO	PIPING CLASS
10"-P-100-085-B2-H	P-112A/B	H-201 inlet manifold	B2 - 300# 5Cr-1/2Mo ASTM A335 P5
DESIGN P / T	OPER. P / T	TEST P (HYDRO)	PIPE WALL
24.0 barg / 380.0 C	18.99 barg / 348.7 C	44.7 barg (1.5xPxST/5=1.24)	SCH 40 (9.27 mm)
INSULATION	PWHT	NDE	STRESS
H 100 mm	YES - all thk (P-No.5B, B31.3 Tbl	100 % RT of BW; 100 % MT/PT of fillets;	CAT 1 - T > 300 C & NPS >= 6; rotating equipment (
P&ID	FLUID / PHASE	PAINT	SPOOLS / WELDS
CFU-100-PR-PID-006, CFU-200-PR-PID-012	P / L Atmospheric residue to H	HT silicone/Al (insul.)	33 spools; 89 shop / 32 field